



Imports, supply chains and firm productivity

Carol Newman^{a,*}, John Rand^b, Finn Tarp^b^a Department of Economics, Trinity College Dublin, Ireland^b Department of Economics, University of Copenhagen, Denmark

ARTICLE INFO

JEL Codes:

D22
F13
F61
F63
F14
O12
O53
L60

Keywords:

Imports
Supply chains
Productivity
Vietnam

ABSTRACT

This paper explores how competition-induced productivity gains from imports in intermediate producing sectors transmit through the supply chain. Based on firm-level panel data from Vietnam, we show that in addition to the productivity premium associated with importing intermediate inputs, firms that use domestically-produced inputs from more import-intensive sectors also have higher productivity. We find evidence that import competition leads to product differentiation, in particular higher quality output in sectors where there is greater scope for quality variation, leading to better quality domestically-produced inputs. We also find evidence that non-importing firms that experience productivity gains due to greater import intensity in upstream sectors change their input mix and become more capital intensive. We conclude that ignoring the gains from trade through this mechanism may significantly underestimate the impact of trade on productivity.

1 Introduction

The impact of foreign competition on productivity is an important policy question for emerging economies undergoing trade liberalization. It is widely understood that import competition is likely to facilitate factor reallocations from the least efficient to more productive firms, a mechanism generally recognized as an important source of aggregate productivity gains from trade (Melitz, 2003).¹ It is also possible that import competition impacts on firm-specific productivity levels both within import-competing sectors, and indirectly through vertical (between-sector) supply chain linkages from upstream to downstream sectors. In this study, we provide new empirical evidence for an emerging economy (Vietnam) on the way in which the import of inputs impacts on firm-level productivity along each of these dimensions.

We consider three specific mechanisms through which import competition can impact firm-level productivity. First, we study the extent to which exposure to import competition impacts on the productivity levels of surviving firms. While we expect the average productivity of a sector to increase in response to import competition due to the reallocation of resources from the least productive firms, which are

likely to exit, to more productive surviving firms, it is also possible that the productivity of the surviving firms increases. It is well established that protection from competition reduces the incentive for firms to reduce slack in their production processes (Amiti and Konings, 2007; Jabbour et al., 2019; Konings, 2008). As such, exposure to greater competition from imports will force firms to reduce slack to survive thereby leading to within-firm productivity improvements.

Second, we consider the extent to which downstream firms that import intermediates benefit from technology spillovers, i.e. variety or quality effects associated with using higher quality imported inputs. This mechanism reflects the idea that in an emerging economy, imports from more advanced economies will be of a higher quality and embody more sophisticated technology than locally-produced inputs. As such, an expansion of imported intermediates will lead to technology diffusion through greater variety, better quality inputs and new technologies embodied in those inputs (Grossman and Helpman, 1991). There is empirical support for this mechanism in a number of country studies (see, for example, Fernandes (2007) (Colombia), Amiti and Konings (2007) (Indonesia), Shor (2004) (Brazil), Topalova and Khandelwal (2011) (India), Kasahara and Rodrigue (2008) (Chile), Goldberg et al.

* Corresponding author.

E-mail addresses: cnewman@tcd.ie (C. Newman), john.rand@econ.ku.dk (J. Rand), finn.tarp@econ.ku.dk (F. Tarp).¹ Exposure to competition will force the least efficient firms to exit or switch sector. Empirical evidence of this mechanism can be found in Tybout et al. (1991) and Pavcnik (2002) for Chile, and Eslava et al. (2004) and Fernandes (2007) for Colombia.

(2010) (India) and Halpern et al. (2015) (Hungary)). However, there are also studies, which suggest that it may not be an important mechanism (see, for example, Van Biesebroeck (2008) (Colombia) and Muendler (2004) (Brazil)).

Third, it is possible that in addition to reducing slack in response to import competition, domestic producers improve the quality of the goods they produce (Antoniades, 2015) or expand product scope (Goldberg et al., 2010) due to the need to compete with imports. If competition from imported intermediates leads to better quality domestically-produced inputs or more domestically-produced varieties, this may lead to productivity improvements in downstream sectors that purchase these inputs.² As such, we explore whether firms that do not import inputs, but purchase them from domestic producers, experience a productivity gain due to import competition in intermediate goods sectors. Melitz and Redding (2014) refer to this as a possible ‘missing’ gain from trade. They highlight the potential for trade to lead to endogenous changes in domestic productivity through gains occurring at each intermediate stage of production. This indirect effect has to our knowledge, not previously been explored empirically.³ As such, our paper contributes to the literature by considering the vertical indirect effects of exposure to inputs whereby the productivity enhancing effects of competition from imported inputs in upstream sectors transmit through the supply chain to downstream final goods producers.

In sum, the following three productivity-enhancing mechanisms are tested:

- Mechanism 1: Impact of expanded imports on domestic competitors – *horizontal effect*
- Mechanism 2: Impact on downstream firms, which import intermediates, from upstream sectors – *vertical direct effects*
- Mechanism 3: Impact on downstream firms, which do not import intermediates, from upstream sectors – *vertical indirect effects*

In other words, we distinguish between the *direct* horizontal effect on firms that compete with imports, the *direct* vertical effect on firms that import intermediates, and the *indirect* effect on firms that use domestically-produced inputs. In addition, we explore the channels underlying the import-induced productivity improvements associated with the third mechanism. Specifically, we study the quality innovation channel by examining how import penetration relates to mark-ups, distinguishing between the cost and price components of mark-ups and the scope for quality differentiation within sectors.

The setting for our study is Vietnam.⁴ This is a particularly relevant case for exploring the impact of trade. This is so, given the wide range of reforms related to trade liberalization over the last three decades since the *Doi Moi* reform process began in 1986. Trade liberalization took the form of the removal of export taxes and non-tariff barriers and the negotiation of a series of trade agreements with ASEAN, the US and the EU, ultimately leading to WTO accession in 2007.⁵ Our data includes an extensive firm-level panel dataset on almost 90,000 manufacturing firms in Vietnam for the period 2008 to 2017. We match the firm-level data to

import and export statistics from COMTRADE at the 4-digit sector level. To measure vertical effects, we use a measure of supply chain linkages that captures the extent of exposure of a sector to imports upstream using Supply Use Tables (SUT) for Vietnam, an approach previously used to identify vertical technology transfers from foreign firms (Javorcik, 2004; Newman et al., 2015).⁶

To empirically identify the three mechanisms through which imports impact on productivity, we build on the de Loecker (2013) approach for detecting productivity improvements following entry into export markets and accommodate the endogenous productivity process associated with firm-level exposure to trade in the estimation procedure. We consider the direct effects through importing and exporting and the indirect effects through exposure to import competition within-sector and through the supply chain.⁷

Our empirical analysis provides evidence for all three proposed mechanisms. We find a positive association between the proportion of imports into a sector and firm-level productivity. We also find that there is a productivity premium for firms that import intermediates. Moreover, we show that an increase in imports into input-producing upstream sectors increases the productivity of *all* downstream firms – both those that import intermediates and those that do not. Through analyzing how import penetration affects firm-specific mark-ups we also provide evidence that product differentiation, most likely related to quality improvements in response to import competition, is a likely factor contributing to the increased productivity of downstream firms that use domestically-produced inputs. Finally, we show that non-importing firms that experience productivity gains associated with greater import intensity in upstream sectors increase the capital intensity of their production process suggesting that imports could have long term effects on productivity and production technology through this channel. Our findings indicate that ignoring the gains from trade through the supply chain may underestimate the impact of trade on the productivity of domestic firms as predicted by Melitz and Redding (2014).

The remainder of the paper is organized as follows. In Section 2 we present our empirical approach. Section 3 introduces the datasets and describes the evolution of the manufacturing sector in Vietnam over the period of analysis. Section 4 presents the results, and Section 5 concludes.

2. Empirical approach

To empirically examine the relationship between importing and productivity we follow de Loecker (2013) and allow for an endogenous productivity process whereby firm-level actions and policy variables that influence firm-level productivity, and therefore input choices, are included in the control function in the estimation of productivity.⁸ We are specifically interested in the impact of importing on productivity and

⁶ While Blalock and Veloso (2007) also use input–output tables to measure supply chain linkages, they focus on backward linkages from downstream import firms to upstream domestic suppliers of inputs.

⁷ Trade liberalization is also likely to affect productivity through an expansion in exports; and productivity gains are possible through improvements in within-firm efficiency induced by exposure to competitive pressures on export markets. Moreover, firms may incur a sunk cost of entry into export markets and so measured productivity improvements may take some time to materialize. A large empirical literature exists providing evidence for these mechanisms. See Bernard and Jensen (1999), Bigsten et al. (2006), Blalock and Gertler (2004), Clerides et al. (1998), Foster et al. (2008), Newman et al. (2016), Roberts and Tybout (1997), and Van Biesebroeck (2005) for examples. While we do not consider these mechanisms here, we do control for the simultaneous impact of exposure to export markets in our estimation of the endogenous productivity process.

⁸ For a comprehensive discussion on productivity measurement and its determinants see Syverson (2011).

² Blalock and Veloso (2007) focused on backward linkages through the supply chain in investigating the impact of imports on the productivity growth of firms in Indonesia. They find evidence that importing is a source of technology transfer for upstream firms supplying import-intensive downstream sectors.

³ Caliendo and Parro (2015) provide support for exploring trade impacts through the supply chain. They studied the importance of input–output linkages in evaluating the trade and welfare effects of tariff reductions, applying a model to identify the effects of NAFTA tariff reductions and find that both the trade and welfare effects of tariff reductions are significantly smaller when analysts ignore input–output linkages.

⁴ A relevant and comprehensive background study is McCaig and Pavcnik (2013).

⁵ See Abbott et al. (2009) for a discussion on the impact of Vietnam’s Trade Agreements on development.

Table 1

Manufacturing firms: descriptive statistics.

	Number of firms	Value Added (million VND)	Employees	Capital (million VND)	Importer (%)	Exporter (%)	Exits (%)
2008	25,129	5,220	107	22,732	10.73	13.98	33.05
2009	25,023	6,442	108	26,990	14.15	16.19	27.85
2010	29,545	6,383	105	26,474	15.32	15.62	33.29
2011	30,327	7,029	112	25,107	16.33	18.79	24.37
2012	33,626	6,763	103	26,232	14.03	16.62	28.74
2013	32,722	7,529	112	29,744	15.02	19.66	25.92
2014	33,586	8,452	116	30,746	15.39	19.84	30.78
2015	33,650	9,206	114	33,453	16.07	20.97	33.55
2016	34,031	10,394	119	36,798	14.75	20.47	21.29
2017	36,544	10,034	111	36,939	14.25	18.85	

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

Note: Value added is deflated using the provincial GDP deflator and capital is adjusted based on a national capital deflator. We obtained both deflators from the General Statistical Office (GSO). Data on importers are not available in 2017 so we impute based on whether the firm imported in the previous period. Results reported are robust to dropping 2017 from the analysis.

so consider three variables that capture the mechanisms we wish to identify: *Mechanism 1*: the extent of import penetration in a sector; *Mechanism 2*: whether the firm imports intermediates; and *Mechanism 3*: the proportion of inputs into the sector that is imported. Assuming an exogenous productivity process would unrealistically imply that past import experience and past exposure to import competition have no effect on direct technological improvements, product innovation or product quality upgrading.⁹ By including these variables in the control function, we allow for the fact that they are endogenous to the productivity process. In other words, in estimating productivity, we allow the firm's exposure to import competition, experience in importing intermediates, and access to upstream imported intermediates to affect productivity through input decisions. We uncover the relationship between each of these factors and productivity by estimating the endogenous productivity process ex post. We also include an indicator for whether the firm is an exporter in the control function. There are two reasons why this is important. First, the endogenous relationship between exporting and productivity is well established (de Loecker, 2013). Second, Vietnam is an important destination for export processing.¹⁰

The main way in which our approach deviates from the standard approach is in the specification of the law of motion of productivity. As in Akerberg et al. (2015) and de Loecker (2013), we assume a Cobb-Douglas production function:

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \omega_{it} + e_{it} \quad (1)$$

where y_{it} is the log of value added, l_{it} is the log of the labor input, k_{it} is the log of the capital input, ω_{it} is unobserved productivity, and e_{it} is an unanticipated shock or random error term. Simultaneity between productivity shocks (observed by the firm but not by the econometrician) and input choices leads to bias in OLS estimates of the coefficients on these inputs in a standard production function.¹¹ The standard approach

⁹ The extent of import competition in a sector and the proportion of imported inputs available in upstream sectors are not firm-level actions. This is, however, information that firms have along with knowledge of their productivity distribution. We do not have information about the firm's productivity distribution or the input choices the firm makes in response to the general level of imports into the sector and into upstream sectors. Not accounting for these factors in estimating productivity will lead to biased estimates.

¹⁰ The Customs Handbook on International Merchandise Trade Statistics of Vietnam 2016 (<https://wtocenter.vn/thong-ke/13146-customs-handbook-on-international-merchandise-trade-statistics-of-vietnam-2011-2017>) reports that as a percentage of merchandise exports, processing exports rose from 47.2 to 76.0 between 2008 and 2015.

¹¹ For example, higher productivity firms may decide to employ more workers, leading to an upward bias in the coefficient on labor if we do not control for productivity. Labor decisions could also be countercyclical; and the coefficient on capital biased for similar reasons.

to estimating the production function parameters in the presence of such bias is to use a semi-parametric estimator, which applies some structure to the underlying decision-making process of firms (for example, Olley and Pakes (1996), Levinsohn and Petrin (2003) and Akerberg et al. (2015)). We assume productivity evolves according to an exogenous first-order Markov process whereby:

$$E(\omega_{it} | \omega_{it-1}, \omega_{it-2}, \dots, \omega_{i1}) = E(\omega_{it} | I_{it-1}) = E(\omega_{it} | \omega_{it-1}) \quad (2)$$

where I_{it-1} is the information set at time $t-1$, and we assume all past realizations of productivity are part of that information set. In other words, the firm expectations about future productivity depend only on the productivity in the previous period. In the standard Olley and Pakes (1996) approach, by assuming that investment, and hence the capital stock is chosen at time $t-1$, productivity can be proxied by a function of investment and capital as in equation (3).¹²

$$\omega_{it} = g(k_{it-1}, i_{it-1}) + \xi_{it} \quad (3)$$

ξ_{it} is a productivity shock, which by definition is uncorrelated with any factors included in the information set of the firm, I_{it-1} . As pointed out by de Loecker (2013), this is problematic when we are interested in identifying the impact of factors on productivity that are endogenous to the productivity process, in our case trade-related variables that capture the three mechanisms through which imports could impact on productivity. If we have any reason to believe that firms will adjust their input choices in response to these trade-related variables then the capital and labor coefficients estimated using the standard Olley and Pakes (1996) approach will be biased. In other words, ξ_{it} in equation (3), and therefore e_{it} in equation (1), will include the productivity effects of trade and too much of the variation in output will be attributed to the variation in inputs.

To correct for this we allow these variables to impact future productivity and so re-specify equation (3) to include the vector, $\mathbf{T}_{it-1}$, which measures the firm's exposure to trade along the various dimensions of interest.

$$\omega_{it} = g(k_{it-1}, i_{it-1}, \mathbf{T}_{it-1}) + \xi_{it} \quad (4)$$

The timing of the choice of labor is also relevant. We follow the standard assumption that labor choice takes place at the same time productivity is realized. This implies that the labor input, determined at time t , correlates with ξ_{it} . This is not so for the lag of labor, since it is part of the information set at time $t-1$ and so it can be used as an instrument for l_{it} (Akerberg et al., 2015).

Combined with the endogenous productivity process in equation (4), we generate a set of moment conditions that can be used to estimate the production function parameters.

¹² Levinsohn and Petrin (2003) use intermediate inputs instead of investment.

Table 2
Exposure to imports.

Sector	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Exposure to imports within sector (per cent)										
15: Food and Bev	23	21	20	21	11	13	12	14	15	16
17: Textiles	46	45	49	49	45	48	45	43	46	47
18: Wearing apparel	64	62	60	61	6	6	6	5	6	5
19: Leather	59	57	57	61	6	7	6	6	10	9
20: Wood	35	30	31	31	18	18	23	20	22	20
21: Paper	8	8	8	7	19	21	20	20	24	22
22: Recorded media	12	10	8	8	4	4	3	4	4	7
24: Chemicals	14	13	14	17	48	48	46	47	51	52
25: Rubber and plastic	23	22	22	22	28	30	30	32	39	39
26: Other non-metallic minerals	18	12	15	15	10	11	10	12	13	13
27: Basic metals	28	16	19	22	56	55	55	58	60	55
28: Fabricated metal	15	14	13	13	20	23	23	30	26	22
29: Machinery and equipment	33	30	34	38	78	77	78	83	84	83
30: Office machinery	50	45	49	55	31	36	37	40	41	38
31: Electrical machinery	39	38	40	42	44	45	46	45	50	52
32: Communication equipment	37	35	29	35	61	59	55	53	54	54
33: Medical instruments	54	57	50	51	65	64	67	68	74	78
34: Motor vehicles	26	19	22	24	28	25	30	34	38	32
35: Other transport	19	11	20	24	10	10	11	14	20	16
36: Furniture and manu n.e.c.	45	41	43	43	10	11	12	12	14	13
Exposure of downstream sectors to upstream imports (per cent)										
15: Food and Bev	40	40	37	38	35	38	38	42	42	40
17: Textiles	24	25	27	25	45	46	46	49	47	45
18: Wearing apparel	26	27	29	28	38	39	39	41	40	39
19: Leather	39	37	39	42	42	43	43	44	42	40
20: Wood	25	23	24	29	45	47	51	48	51	41
21: Paper	12	12	12	13	28	29	30	30	29	28
22: Recorded media	13	12	13	13	27	29	29	30	29	28
24: Chemicals	21	22	24	25	50	51	50	54	51	52
25: Rubber and plastic	30	26	28	26	59	60	60	61	60	59
26: Other non-metallic minerals	22	21	16	18	14	15	14	24	18	22
27: Basic metals	20	12	13	16	36	39	39	44	38	34
28: Fabricated metal	19	17	16	18	38	41	41	46	40	37
29: Machinery and equipment	20	21	21	23	45	48	49	54	50	47
30: Office machinery	28	27	24	33	61	63	60	61	62	60
31: Electrical machinery	19	19	17	19	38	39	39	42	40	37
32: Communication equipment	22	23	21	26	48	50	48	50	50	49
33: Medical instruments	26	28	24	26	42	43	43	46	45	45
34: Motor vehicles	18	17	17	18	27	29	28	32	30	28
35: Other transport	14	11	13	15	28	29	28	31	28	25
36: Furniture and manu n.e.c.	24	22	23	26	37	39	42	41	41	35

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

Note: Exposure is the share of imported input into sectors. Abbreviated names of sectors. Full names are in Table A1 of the Appendix.

$$E \left[\xi_{it}(\beta_l, \beta_k) \begin{pmatrix} l_{it-1} \\ k_{it} \end{pmatrix} \right] = 0 \quad (5)$$

To obtain $\xi_{it}(\beta_l, \beta_k)$ we perform a non-parametric regression of $\omega_{it}(\beta_l, \beta_k)$ on $\omega_{it-1}(\beta_l, \beta_k)$ where $\omega_{it}(\beta_l, \beta_k) = \hat{\phi}_{it} - \beta_l l_{it} - \beta_k k_{it}$ and $\hat{\phi}_{it}$ is the predicted output from a regression of actual output, y_{it} , on the inputs, the proxy variables (investment and capital) and the vector of trade variables.

Having estimated the production function parameters, we back out the unbiased estimate of firm-level productivity. Next, we identify the impact of the trade variables on productivity through an ex-post parametric estimation of the underlying productivity process as described in equation (6).¹³

¹³ Our approach to estimating the relationship between productivity and trade allows the evolution of a firm's productivity to depend on past trade experiences, thus allowing productivity to evolve endogenously. This approach, pioneered by de Loecker (2013), addresses many of the empirical concerns associated with assuming an exogenous productivity process, which does not allow trade exposure to influence future innovations or improvements in technology. One drawback of our approach, however, is that there may be other factors, correlated with a firm's trade experience that are also correlated with productivity growth, that confound this relationship. We control for a range of sector and time fixed effects in our analysis to address this concern. Nevertheless, this caveat should be borne in mind when interpreting our results.

$$\hat{\omega}_{it}(\hat{\beta}_l, \hat{\beta}_k) = g(\hat{\omega}_{it-1}(\hat{\beta}_l, \hat{\beta}_k), \mathbf{T}_{it-1}) + u_{it} \quad (6)$$

where the functional form of $g()$ depends on the form of non-parametric regression used to obtain $\xi_{it}(\beta_l, \beta_k)$. We use a third-order polynomial for $g()$ but our results are robust to alternative specifications. To examine the impact of the trade variables on productivity we estimate $g()$ parametrically and use the coefficients to compute the marginal effect of import penetration, directly importing and the proportion of imports from upstream sectors on productivity to test the three different mechanisms through which trade can influence firm productivity as described in the introduction.¹⁴

3. Data

Our main data source is the Vietnam Enterprise Survey (VES) collected annually by the General Statistics Office (GSO). This dataset

¹⁴ The marginal effects are described in the Appendix after Table A3 where we present the results of the ex-post parametric estimation of the productivity process.

Table 3
Production Function Estimation.

Sector	$\hat{\beta}_l$	$\hat{\beta}_k$
1: Food, Beverages		
Exogenous productivity process	0.855	0.327
Endogenous productivity process	0.839	0.315
2: Textiles, Wearing apparel, Leather		
Exogenous productivity process	0.951	0.155
Endogenous productivity process	0.909	0.191
3: Wood, Paper, Recorded media		
Exogenous productivity process	0.996	0.193
Endogenous productivity process	0.928	0.212
4: Recorded media		
Exogenous productivity process	0.966	0.241
Endogenous productivity process	0.983	0.208
5: Chemicals		
Exogenous productivity process	0.758	0.195
Endogenous productivity process	0.774	0.423
6: Rubber and other non-metallic minerals		
Exogenous productivity process	0.863	0.301
Endogenous productivity process	0.870	0.273
7: Basic metals, Fabricated metals		
Exogenous productivity process	0.934	0.256
Endogenous productivity process	0.943	0.205
8: Machinery and equipment		
Exogenous productivity process	0.959	0.171
Endogenous productivity process	0.838	0.290
9: Motor vehicles, Other transport		
Exogenous productivity process	0.860	0.342
Endogenous productivity process	0.867	0.285
10: Furniture, Other manufacturing		
Exogenous productivity process	0.969	0.156
Endogenous productivity process	0.920	0.176

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

includes the population of all registered enterprises with 30 employees or more and a representative sample of smaller firms for the years 2008 to 2017.¹⁵ Our total sample after data cleaning includes 314,183 observations on 89,767 manufacturing enterprises. Descriptive statistics of the main firm-specific variables of interest are in Table 1. The dynamic nature of the Vietnamese economy is evident. Between 20 and 33 per cent of firms exited each year; and the proportion of firms that import intermediates ranged from 10 to 16 per cent. The proportion of firms that exported ranged between 13 and 21 per cent.

The measure of imports into a sector are taken from the UN COMTRADE database available through World Integrated Trade Solutions (WITS) and is measured as the value of imports for 4-digit ISIC sectors for Vietnam from the rest of the world. We use this to compute a measure of import penetration at the sector level described in equation (7).

$$propimp_{jt} = \frac{imp_{jt}}{out_{jt} + imp_{jt}} \quad (7)$$

where imp_{jt} is the level of imports into sector j in year t and out_{jt} is the total output (revenue) in sector j in year t .

To measure supply chain linkages, we use the Supply-Use Tables (SUT) for Vietnam in 2012. The SUT maps the supply and use of 164 commodities and services across production activities. We link these production activities to the 4-digit ISIC codes in the VES to produce 122 comparable sector codes, 56 in the manufacturing sector. We use the SUT data to construct a set of weights that captures upstream linkages between sectors, whereby for each sector j , their link with upstream sector u is the proportional contribution of output from sector u to its total input base. We subsequently use these weights to compute a weighted average of imports from upstream sectors as a proportion of

¹⁵ We exclude firms with less than five employees and trim the 99th percentile of the distribution of value added and capital. For more details about the Vietnam Enterprise Survey (VES), see Newman et al. (2013, 2015).

Table 4
Marginal effects of trade on productivity.

	(1) Endogenous	(2) Exogenous
Proportion of imports in total output (Propimp)	0.062*** (0.000)	0.042*** (0.000)
Importer	0.095*** (0.000)	0.004*** (0.008)
Proportion of imported inputs (Propimp_up)	0.078*** (0.000)	0.050*** (0.000)
Proportion of imports in total output for non-importing firms	0.068*** (0.000)	
Differential effect of proportion of imports in total output for importers	−0.039 (0.425)	
Proportion of imports from upstream sectors for non-importing firms	0.054*** (0.000)	
Differential effect of proportion of imports from upstream sectors for importers	0.146*** (0.000)	
Exporter	0.089*** (0.000)	0.014*** (0.000)
Observations	196,838	196,838

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

Note: In column (1) we present the marginal effects of each variable on productivity using the relevant coefficient estimates presented in Table A3 as described in the Appendix. P-values for F-tests of significance of coefficients related to each particular variable are also presented in parentheses in column (1). Column (2) presents the results of a regression of the productivity measure assuming the exogenous productivity process on the four main trade variables. P-values from the t-test of the statistical significance of each of the parameters are also presented in parentheses in column (2). All specifications control for sector $\times$ year fixed effects. *** indicates significance at the 1% level, ** the 5% level and * the 10% level.

total output in the sector imported and produced in Vietnam,¹⁶ as described in equation (8).

$$primp_up_{jt} = \sum_{u=1}^{J-1} w_{ujt} \left(\frac{imp_{ut}}{out_{ut} + imp_{ut}} \right) \quad (8)$$

Table 2 presents the (average) extent of exposure of each 2-digit sector to imports, as measured by equation (7), and to imported inputs, as measured by equation (8).¹⁷ There is a lot of variation over time and across sectors. In most sectors, the proportion of imports into the sector and the proportion of imported inputs increased. Some declines are evident, however, reflecting the combination of growth in domestically-produced output over this period and the world financial crisis, which meant that more domestic output remained in-country.¹⁸

4. Results

We estimate production functions separately for 10 sub-sectors of manufacturing.¹⁹ The labor and capital elasticities for two different

¹⁶ Calculated as imports into the sector/(output produced domestically in the sector + imports into the sector). Imported inputs from the firms' own sector are also included in this measure.

¹⁷ Vietnam uses an industry classification code that is similar to ISIC Rev3. We use abbreviated descriptions for the purpose of exposition. The list of sectors corresponding to each 2-digit code with full definition is provided in Table A1 of the Appendix.

¹⁸ Given the potential for the collapse in trade during the financial crisis to impact our results we estimate all specifications excluding 2007–2009, and all of the results hold (see Tables A4–A6 in the Appendix).

¹⁹ 2-digit sectors are grouped together to estimate productivity. The groupings used are given in Table 3.

Table 5
Determinants of positive import effect.

	(1) Propimp	(2) Propimp_up
Lag productivity	−0.007** (0.034)	−0.015** (0.007)
Lag medium sized firm	−0.001 (0.001)	0.010*** (0.001)
Lag large firm	−0.010*** (0.003)	−0.002 (0.010)
Lag import firm	−0.023*** (0.003)	0.072*** (0.005)
Lag export firm	0.048*** (0.006)	−0.017 (0.012)
Lag foreign owned firm	0.007** (0.003)	0.023*** (0.004)
Lag state owned firm	0.005** (0.002)	0.002 (0.004)
Lag HHI	0.117** (0.055)	−0.016 (0.027)
Lag FR	−0.041*** (0.015)	0.040*** (0.010)
Lag SR	−0.047*** (0.010)	0.070*** (0.023)
Sector size	0.004 (0.004)	0.002 (0.003)
R-squared	0.558	0.563
Observations	179,809	179,809

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

Note: All specifications control for 4-digit sector fixed effects, year fixed effects and 2-digit sector $\times$ year fixed effects. Standard errors clustered at the sector-year level are presented in parentheses. *** indicates significance at the 1% level, ** the 5% level and * the 10% level.

specifications are presented in Table 3: (i) a specification which assumes an exogenous productivity process, as in Olley and Pakes (1996); and (ii) our more flexible specification, which allows for an endogenous productivity process, related to the trade variables.

Adjusting for the endogeneity of trade-related variables in the estimation of productivity has different impacts on the coefficients on labor and capital across sectors. This suggests that in some cases the trade variables correlate positively with firm size and capital stock while in others they correlate negatively.²⁰ The growth in productivity for each sub-sector for the 2008–2017 period for the flexible specification is presented in Table A2 of the Appendix.²¹

To examine the mechanisms through which imports impact on productivity we use the productivity estimates backed out from the elasticity estimates obtained through the estimation of the production function that addresses the endogeneity of the trade-related variables, to examine the relationship between the trade variables and productivity ex-post. We use a third order polynomial in the variables and include all combinations of interactions between each variable. We also control for sector-time fixed effects to address the fact that productivity is estimated separately for different sub-sectors. The results of this regression are presented in Table A3 of the Appendix. We use these coefficients to aggregate the marginal effects of each of the trade variables of interest on productivity allowing us to test the three main mechanisms described in the introduction. The equations used to compute the marginal effects are provided after Table A3 in the Appendix. The results for the marginal effects are presented in column (1) of Table 4. We use an F-test of the

²⁰ It could be, for example, that in some sectors firms contract when exposed to more import competition. In other sectors it could be that entry into export markets leads to an expansion in labor and capital.

²¹ Average productivity levels across sectors are not comparable given that we estimate production functions separately for each sector. This means we are assuming that firms within sectors share a common technology but that this technology is different between sectors. The productivity trajectories in each sector are comparable.

Table 6
Imports and mark-ups.

	(1)	(2)	(3)	(4)
Prop imports	0.046*** (0.012)	0.036*** (0.012)	0.024 (0.015)	0.034** (0.017)
Productivity		0.438*** (0.015)	0.438*** (0.015)	0.427*** (0.017)
Prop imports * RD			0.452 (0.402)	0.019 (0.395)
Prop imports * HHI				−0.342*** (0.140)
Prop imports * RD * HHI				11.782*** (4.105)
Productivity * HHI				0.293 (0.218)
HHI				−0.424 (0.345)
Labor and Capital	Yes	Yes	Yes	Yes
4-digit sector fixed effects	Yes	Yes	Yes	Yes
Time dummies	Yes	Yes	Yes	Yes
2-digit sector $\times$ time dummies	Yes	Yes	Yes	Yes
R-squared	0.078	0.107	0.107	0.107
N	183.997	183.997	183.997	183.997

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

Note: Standard errors clustered at the sector-year level are presented in parentheses. *** indicates significance at the 1% level, ** the 5% level and * the 10% level.

joint significance of the parameters associated with the marginal effect of interest in the full specification to determine the statistical significance of the estimates. For comparison purposes, in column (2) we present the marginal effects of the trade variables on productivity estimated using the exogenous productivity process, which does not account for the endogenous relationship between trade and the firm's input choices.

The first mechanism we consider is the impact of the level of imports into a sector on the productivity of domestic competitors (*Mechanism 1 - horizontal effects*), measured by the overall effect of the proportion of imports in total output (equation (7)) on productivity. We find a positive and statistically significant association between imports and productivity. A one percentage point increase in the proportion of imports in total output would lead to a 0.062 per cent increase in productivity. During the sample period, the average proportion of imports in total output increased almost 20 percentage points (from 46 per cent in 2007 to 66 per cent in 2017).²² Considering that over the same period average productivity grew by around 30 per cent, the contribution (all else equal) of the first mechanism to average productivity growth would be approximately 4 per cent, which is economically meaningful.

The second and third mechanisms relate to the impact of imports through the supply chain. The second mechanism relates to the direct effect of importing intermediates and is captured by the marginal effect of being an importing firm on productivity. We find that, consistent with the literature, importing inputs is associated with a productivity premium of 0.095 per cent.²³

The third mechanism is concerned with the impact of an expansion in imported intermediates on downstream firms, which do not import intermediates from upstream sectors. To test this mechanism, we consider how the proportion of imports into upstream input-supply sectors impacts on the productivity of downstream firms. We find a positive and

²² Assuming domestic output is constant, a one percentage point increase in the proportion of imports in total output is equivalent to a relative increase in imports of around 1.15 billion USD in value terms over the period of analysis. Average imports into the sectors represented in our data amounted to around 198 billion USD in 2017.

²³ For other evidence of productivity gains from importing see, for example, Shor (2002), Kasahara and Rodrigue (2008), Goldberg et al. (2010), Topalova and Khandelwal (2011), and Halpern et al. (2015), among others.

Table 7

Productivity gains from imports and changes to the input mix.

	(1) Growth in employment	(2) Growth in capital	(3) Growth in capital-labor ratio
Lagged indicator variables:			
Productivity gains from within-sector expansion in imports	−0.002 (0.010)	0.034 (0.041)	0.006 (0.044)
Productivity gains from importing inputs	−0.106 (0.076)	0.109* (0.060)	0.142** (0.069)
Productivity gains from expansion in imports in upstream sector	−0.073*** (0.018)	0.259*** (0.059)	0.286*** (0.066)
Firm fixed effects	Yes	Yes	Yes
Firm and sector time-varying controls	Yes	Yes	Yes
4-digit sector fixed effects	Yes	Yes	Yes
Time dummies	Yes	Yes	Yes
2-digit sector × time dummies	Yes	Yes	Yes
N	121,164	121,164	121,164

Source: Authors' calculations based on Vietnamese Enterprise Surveys.

Note: Standard errors clustered at the sector-year level are presented in parentheses. *** indicates significance at the 1% level, ** the 5% level and * the 10% level.

statistically significant impact of the proportion of imported inputs into upstream sectors on the productivity of downstream firms. The result indicates that a one percentage point increase in the proportion of imported inputs into an upstream sector is associated with a 0.078 per cent increase in productivity of downstream firms. This is slightly higher than the direct impact of imports. To disentangle the extent to which improvements in productivity for firms that directly import intermediates drive this vertical effect, we disaggregate the effect between firms that import intermediates and those that do not. This will give us some indication as to whether the impact is a further direct effect (*Mechanism 2*) or an indirect effect (*Mechanism 3*). The flexible specification presented in Table A3 of the Appendix allows us to examine the differential effect of the proportion of upstream inputs that are imported for non-import and import firms. From the marginal effects presented in Table 4 we see that the effect for non-import firms is positive and statistically significant suggesting that for non-import firms a one percentage point increase in the proportion of imported inputs increases productivity by 0.05 per cent. The differential effects for import firms is significantly higher as might be expected given that it is the firms that directly import inputs that benefit the most from an expansion in the availability of foreign varieties. We conclude that while import firms benefit from an increase in imports into upstream sectors so do non-import firms thus providing evidence for *Mechanism 3*, i.e. productivity gains due to competition from imports in upstream sectors transmitted through the supply chain as proposed by Melitz and Redding (2014). This effect on productivity that comes through the supply chain is an additional gain from trade, which is generally not considered.²⁴

To understand better the mechanisms at work, we explore whether or not there is heterogeneity in the way in which importing impacts on productivity. Since our approach allows us to compute a firm-specific marginal effect for each time period we can use these marginal effects to explore what firm or sector characteristics are associated with these gains from trade. In Table 5, we present the results of two regressions exploring the firm and sector characteristics associated with the marginal effects of the proportion of imports into a sector on productivity

(column (1)) and the proportion of imported inputs into a sector on productivity (column (2)). The firm characteristics we consider are firm productivity in the previous period, dummy indicators for whether the firm is medium sized (between 50 and 299 employees) or large in size (300 employees or more), whether the firm imports intermediates, whether the firm exports, and whether the firm is foreign or state owned. The sector-level variables included are computed at the 4-digit sector level and include the level of industry concentration (measured by the standard Herfindahl-Hirschman Index (HHI)), the concentration of foreign-owned firms measured as the proportion of output produced by foreign-owned firms, the concentration of state-owned firms measured as the proportion of output produced by state-owned firms and the average size of firms in the sector in terms of number of employees. All variables are included in lagged form, and each specification includes 4-digit sector and time fixed effects in addition to 2-digit sector time trends.

As shown in column (1), lower productivity firms benefit more from import competition suggesting that competition from imports may force firms that are not productive to cut slack and become more efficient. Consistent with this, we find that gains are higher in more concentrated sectors, which are likely to have greater scope for reducing slack, and are lower in sectors with a greater concentration of state-owned and foreign-enterprises, which are likely to be more competitive and have less scope for cutting slack. We also find that smaller firms benefit more than larger firms as do foreign-owned and state-owned firms.

Column (2) shows that the impact of an expansion in the proportion of imported inputs is also greater for lower productivity firms. This is not surprising given that higher productivity firms are likely to already be importing intermediates giving them a productivity advantage. The effect is greater for medium-sized firms than for small or large firms. This could be due to smaller firms having trouble accessing imported imports and larger firms already importing intermediates. Consistent with our findings in Table 4, import firms benefit more from upstream imports than non-import firms. The productivity gains from an expansion in imported intermediates is greater in sectors where there is a greater concentration of foreign and state-owned firms suggesting that in these sectors imported intermediates are more accessible. Indeed, foreign-owned firms benefit the most from an expansion of imports into upstream sectors.

In the next part of our analysis, we explore the mechanism through which import competition upstream could increase firm-level productivity in downstream sectors by considering how imports impact on firm-level markups. There are two ways in which mark-ups are likely to be affected by import competition. First, if import competition increases productivity/lowers marginal costs as suggested by the results presented in Table 5, more productive firms can charge higher mark-ups as they can undercut competitors.²⁵ This is unlikely to affect the productivity of downstream firms as it will be reflected in the price of those inputs. Second, more technologically advanced firms could respond to import competition by innovating to improve product quality (Antoniades, 2015) or provide new varieties (Goldberg et al., 2010). If this is the case, then import competition could improve the quality of domestically-produced upstream inputs, or the technology associated with them, which could contribute to productivity gains for firms using domestically-produced inputs downstream. Innovations in product quality will also be reflected in differentiation in mark-ups.²⁶ By examining the relationship between mark-ups and imports, we can separate out the productivity/marginal cost effect of imports on mark-

²⁵ In their seminal theoretical work, Melitz and Ottaviano (2008) show this in the context of export firms.

²⁶ Khandelwal (2010) and Kugler and Verhoogen (2012) show that prices are a good proxy for quality, at least in some industries.

²⁴ Consistent with a large empirical literature estimating the firm-specific productivity gains associated with trade, we also find that in the case of Vietnam, exporting is associated with 9 per cent higher productivity.

ups and shed some light on whether other sources of differentiation are possible mechanisms.²⁷

We estimate the firm-specific mark-ups using the [de Loecker and Warzynski \(2012\)](#) approach as our production function framework as in equation (9).²⁸ This is the ratio between the output elasticity of labor, estimated using the production functions for each sector, and the expenditure share of the wage bill in the total revenue of the firm.

$$MU_{ijt} = \beta_{ij} (\theta_{ijt})^{-1} \quad (9)$$

where β_{ij} is the elasticity of output with respect to labor, specific to each sector; and θ_{ijt} is the expenditure share of wages in total revenue.

To examine the relationship between imports and mark-ups we use the specification given in equation (10).

$$\ln MU_{ijt} = \delta_0 + \delta_1 \text{propimp}_{jt-1} + \gamma \mathbf{X}_{ijt} + \epsilon_{ijt} \quad (10)$$

where propimp_{jt-1} is the extent of import penetration into the sector in the previous period and $\mathbf{X}_{ijt}$ includes controls for labor and capital to account for differences in factor intensity, 4-digit sector fixed effects, time dummies and (2-digit) sector-year interactions to capture general industry-specific trends in mark-ups ([de Loecker and Warzynski, 2012](#)).²⁹ If δ_1 is positive, it indicates that mark-ups are higher in sectors exposed to more import competition. Higher firm-specific mark-ups stem from two possible sources: i) lower marginal costs due to the reallocation of resources to more productive firms which can charge higher mark-ups (this will be reflected in the price and is unlikely to impact on downstream productivity); and ii) a greater ability to charge higher prices due to some other factors that allow firms to differentiate their products (a possible candidate for downstream productivity improvements). If we assume, as in [de Loecker and Warzynski \(2012\)](#), that productivity completely captures differences in marginal costs; we can isolate the price effect of import competition by including a control for productivity in this specification (equation (11)).

$$\ln MU_{ijt} = \delta_0 + \delta_1 \text{propimp}_{jt-1} + \delta_2 \hat{\omega}_{ijt} + \gamma \mathbf{X}_{ijt} + \epsilon_{ijt} \quad (11)$$

where $\hat{\omega}_{ijt}$ is firm-specific productivity, estimated using the approach outlined in [Section 2](#). If the coefficient on import penetration remains positive it suggests that imports are associated with some other form of product differentiation that allows firms to charge higher mark-ups.

Other forms of differentiation could include, for example, quality innovations, another possible source for productivity gains observed for downstream firms that use more inputs from import intensive sectors. [Fan et al. \(2015\)](#) show that a reduction in import tariffs leads to an increase in the quality of goods produced in sectors where there is scope for quality differentiation. To explore this mechanism further, we use a measure of the scope for quality differentiation proposed by [Sutton \(1998\)](#) and used by [Kugler and Verhoogen \(2012\)](#) in examining the relationship between quality, prices and plant size in the case of Colombia. We calculate the measure as the ratio of industry level R&D and advertising expenditures to sales. Since we do not have data on R&D and advertising expenditures for Vietnam, we use the measures for US industries provided in [Kugler and Verhoogen \(2012\)](#) which we match to the Vietnamese 4-digit industry classification. By interacting this measure with sector level imports, we can examine whether import competition leads to higher mark-ups in sectors with more scope for

quality differentiation. The full specification is given in equation (12).³⁰

$$\ln MU_{ijt} = \delta_0 + \delta_1 \text{propimp}_{jt-1} + \delta_2 \hat{\omega}_{ijt} + \delta_4 \text{propimp}_{jt-1} \times RD_j + \gamma \mathbf{X}_{ijt} + \epsilon_{ijt} \quad (12)$$

where RD_j is the index for scope for quality differentiation in a sector. It ranges in value from zero to one, with values closer to one indicating greater scope for quality differentiation within the sector. If quality innovations associated with import competition are a source of variation in mark-ups we would expect δ_4 to be positive.

The results for the specifications in equations (10), (11) and (12) are presented in columns (1) to (3) of [Table 6](#). In column (1), we find a positive association between import penetration and firm-specific mark-ups.³¹ As pointed out above, the positive correlation could be due to lower marginal costs allowing firms to charge higher mark-ups or a greater ability to charge higher prices due to some other factors that allow firms to differentiate their products. Including a control for productivity in column (2), which we assume completely captures differences in marginal costs; we can isolate the price effect of import competition from other differentiating factors. The results show that when we control for productivity (cost variation), the magnitude of the coefficient on import penetration declines somewhat. The coefficient on the productivity term is 0.44 suggesting that productivity/marginal cost differences explain a large part of the variation in mark-ups. The sources of variation in mark-ups related to other differentiating factors (such as quality innovations or new varieties) however still play a non-negligible role.

We extend the analysis to include an interaction term between the measure of scope for quality differentiation and the level of imports. The results are presented in column (3). The coefficient on the interaction term is positive but is not statistically significant suggesting that there is no difference in the impact of import penetration on mark-ups in sectors with more scope for quality differentiation. The extent to which there is scope for such differentiation will depend, however on the sector-level concentration. In more concentrated sectors, where there is less competition, there is potentially more scope for product differentiation in response to competitive pressures from imports. To explore this possibility, we include an interaction term between imports and productivity, and the 4-digit sector-level Herfindahl-Hirschman Index (HHI).

The results are presented in column (4) of [Table 6](#). The coefficient on the triple interaction term is positive and statistically significant. This suggests that there is heterogeneity in the association between import competition and mark-ups in more concentrated sectors where there is greater scope for both price and quality differentiation.

In the final part of our analysis, we examine the extent to which the productivity gains from imports may be associated with other changes at the firm level, in particular, input usage. We construct indicator variables that take a value of one for firms that experience productivity gains due to an expansion in imports through each of the three channels. We then consider how the lag of these indicator variables relate to changes in the input usage of firms. We consider three different measures of input usage: (1) employment growth; (2) growth in fixed capital; and (3) growth in the capital-labor ratio. The results of a fixed effects regression including the same set of control variables as in [Tables 5 and 6](#) are in [Table 7](#).³²

Firms that experience productivity gains from an expansion in

²⁷ [Edmond et al. \(2015\)](#) also examine the impact of trade on mark-ups and find that trade reduces mark-up distortions in the case of Taiwanese manufacturing.

²⁸ We do not observe physical output in our data but use deflated revenue in its place. [de Loecker and Warzynski \(2012\)](#) highlight that even in the absence of data on physical output, their approach is still informative about the correlation between mark-ups and firm/sector characteristics, which is the purpose of this exercise.

²⁹ Mark-ups are included in log form.

³⁰ The 4-digit industry fixed effects absorbs RD_j .

³¹ This is consistent with [Brandt et al. \(2017\)](#) who find a negative relationship between input tariffs and markups in the case of China.

³² In this specification, we are interested in isolating the within-firm effects of productivity improvements associated with each import channel and the firm-level changes in input usage. As such we include firm fixed-effects to control for time-invariant firm-specific characteristics that could influence (lagged) productivity improvements associated with imports and changes in input usage.

imports into their own sector (*Mechanism 1*), do not appear to change the level of labor or capital inputs, or the capital-labor ratio. This suggests that productivity gains through this channel do not impact on the type of technology that firms use in their production of output. There is some evidence that firms that experience productivity gains from importing inputs directly (*Mechanism 2*) increase their level of capital and therefore their capital-labor ratio. The largest effects, however, are on the composition of inputs used by firms, which experience productivity gains associated with an expansion of imports into upstream sectors more generally (*Mechanism 3*). They reduce the numbers they employ, use more capital, and increase their capital-labor ratio by almost 30 per cent. We do not have data to unpack these results. However, we note that the effect of productivity gains from importing on the production technology of firms is associated with productivity gains coming through the supply chain (rather than within-sector competition from imports). This suggests that an expansion in imports upstream may lead to greater availability of capital goods (for example, better machinery) downstream, which leads to firms using more capital and less labor inputs. We cannot rule out other possible explanations. For example, the observed changes in the input mix could be related to firm-level investments in R&D and innovation, made possible by the productivity gains made in the previous period. However, we only observe the effects for firms that experience productivity gains due to an expansion of imports into upstream sectors (or through importing directly), providing support for the former explanation. Future work is needed to fully understand the potential channels at work.

Overall, our results show that in addition to productivity gains associated with directly importing intermediates, using domestically-produced inputs from more import-intensive sectors leads to higher levels of productivity. This suggests, as highlighted by Melitz and Redding (2014), that a reorganization of production in upstream sectors in response to import competition can have productivity-enhancing effects for all downstream firms. Moreover, we find some evidence that the channel through which import competition upstream impacts on downstream firms that use domestically-produced inputs is through its impact on product differentiation, such as quality improvements, in upstream sectors, which lead to productivity-enhancing effects for all downstream firms. Finally, we find some evidence that productivity gains from an expansion in imports into upstream sectors is associated with a shift towards more capital-intensive production by downstream firms. This suggests that in addition to leading to productivity improvements, imports may indirectly alter the production technologies of non-importing firms downstream. The implication is that this previously underexplored channel could have even larger effects on productivity over the longer term.

5. Conclusion

In this paper, we explored the relationship between imports and the productivity of firms with the aim of disentangling three mechanisms through which productivity gains may come about. We investigated both within-sector or horizontal effects and between-sector or vertical effects of imports through the supply chain. We used firm-level manufacturing enterprise data covering almost 90,000 firms from Vietnam for the 2008 to 2017 period, matched with both trade data from COMTRADE and input-output tables to measure supply chain linkages.

We found evidence that imports impact on the productivity of domestic firms, for competitors, for firms directly importing intermediates, and for firms in downstream sectors through the supply chain. We found suggestive evidence that product differentiation in response to import competition, most likely related to quality improvements at least in sectors where there is scope for quality differentiation, is one possible mechanism through which upstream import competition benefits non-importing downstream firms. We also found evidence that the firms experiencing productivity gains through this channel change their input mix and become more capital intensive. Future work is needed to further explore this mechanism.

Overall, our results show that there are spillover effects of the productivity gains experienced by upstream domestic firms for downstream firms that use these inputs. These spillovers are of a large magnitude. Accordingly, our findings highlight that previous empirical studies may have underestimated the productivity gains from trade. We believe this supports with novel empirical evidence the theoretical case made by Melitz and Redding (2014), and it certainly suggests that this line of research should attract attention in both further empirical work and in practical policy.

CRedit authorship contribution statement

Carol Newman: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **John Rand:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Finn Tarp:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

This study began within the UNU-WIDER project on ‘Structural Transformation and Inclusive Growth in Vietnam’ funded through income from an endowment fund with additional contributions to the work programme from Denmark, Finland, Sweden, and the United Kingdom and a special contribution from the University of Copenhagen (see site <https://www.wider.unu.edu/project/structural-transformation-and-inclusive-growth-viet-nam>). The authors are grateful for long-term collaboration with staff at UNU-WIDER, the Central Institute of Economic Management (CIEM) and the General Statistics Office (GSO) in Hanoi, Vietnam. Our thanks also go to the editor and two anonymous reviewers for their comments and suggestions, which greatly improved the paper. We also thank participants at various conferences and seminars where we presented this paper including Trinity College Dublin, the University of Copenhagen, the University of Nottingham, and the Nordic Conference in Development Economics in Norway, the European Economics Association Conference, the International Economics Association Conference, and the Irish Economics Association Conference. We have also benefitted from comments from Miklós Koren, Chris Udry and Gaia Narciso. The usual disclaimer applies.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.worlddev.2023.106371>.

References

- Abbott, P., Bentzen, J. S., & Tarp, F. (2009). Trade and development: Lessons from Vietnam's Past Trade Agreements. *World Development*, 37, 341–353.
- Akerberg, D., Caves, K., & Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6), 2411–2451.
- Amiti, M., & Konings, J. (2007). Trade liberalization, intermediate inputs and productivity: Evidence from Indonesia. *American Economic Review*, 97, 1611–1638.
- Antoniades, A. (2015). Heterogeneous firms, quality and trade. *Journal of International Economics*, 95, 263–273.
- Bernard, A. B., & Jensen, J. B. (1999). Exceptional exporter performance: Cause, effect or both? *Journal of International Economics*, 47, 1–25.

- Bigsten, A., Collier, P., Dercon, S., Fafchamps, M., Gauthier, B., Gunning, J. W., ... Zeufack, A. (2006). Do African manufacturing firms learn from exporting? *Journal of Development Studies*, 40, 115–141.
- Blalock, G., & Gertler, P. J. (2004). Learning from exporting revisited in a less developed setting. *Journal of Development Economics*, 75, 397–416.
- Blalock, G., & Veloso, F. M. (2007). Imports, productivity growth, and supply chain learning. *World Development*, 35, 1134–1151.
- Brandt, L., Van Biesebroeck, J., Wang, L., & Zhang, Y. (2017). WTO accession and performance of Chinese manufacturing firms. *American Economic Review*, 107(9), 2784–2820.
- Caliendo, L., & Parro, F. (2015). Estimates of the trade and welfare effects of NAFTA. *Review of Economic Studies*, 82(1), 1–44.
- Clerides, S. K., Lach, S., & Tybout, J. R. (1998). Is learning by exporting important? Micro-dynamic evidence from Columbia, Mexico and Morocco. *Quarterly Journal of Economics*, 113, 903–947.
- de Loecker, J. (2013). Detecting learning by exporting. *American Economic Journal: Microeconomics*, 5(3), 1–21.
- de Loecker, J., & Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471.
- Edmond, C., Midrigan, V., & Xu, D. (2015). Competition, markups and the gains from international trade. *American Economic Review*, 105(10), 3183–3221.
- Eslava, M., Haltiwanger, J., Kugler, A., & Kugler, M. (2004). The effects of structural reforms on productivity and profitability enhancing reallocation: Evidence from Columbia. *Journal of Development Economics*, 75, 333–371.
- Fan, H., Li, Y., & Yeaple, S. (2015). Trade liberalization, quality and export prices. *Review of Economics and Statistics*, 97(5), 1033–1051.
- Fernandes, A. M. (2007). Trade policy, trade volumes and plant-level productivity in Columbian manufacturing industries. *Journal of International Economics*, 71, 52–71.
- Foster, L., Haltiwanger, J., & Syverson, C. (2008). Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability? *American Economic Review*, 98, 394–425.
- Goldberg, P., Khandelwal, A., Pavcnik, N., & Topalova, P. (2010). Imported intermediate inputs and domestic product growth: Evidence from India. *Quarterly Journal of Economics*, 125(4), 1727–1767.
- Grossman, G. M., & Helpman, E. (1991). *Innovation and growth in the global economy*. Cambridge, MA: MIT Press.
- Halpern, L., Koren, M., & Szeidl, A. (2015). Imports and productivity. *American Economic Review*, 105, 3360–3703.
- Jabbour, L., Zhigang, T., Vanino, E., & Zhang, Y. (2019). The good, the bad and the ugly: Chinese imports, European Union anti-dumping measures and firm performance. *Journal of International Economics*, 65(1), 151–165.
- Javorcik, B. S. (2004). Does Foreign Direct Investment Increase the Productivity of Domestic Firms? In Search of Spillovers Through Backward Linkages. *American Economic Review*, 94, 605–632.
- Kasahara, H., & Rodrigue, J. (2008). Does the use of imported intermediates increase productivity? Plant-level evidence. *Journal of Development Economics*, 87, 106–118.
- Konings, J. (2008). Heterogeneous responses of firms to trade protection. *Journal of International Economics*, 76(2), 371–383.
- Kugler, M., & Verhoogen, E. (2012). Prices, plant size, and product quality. *Review of Economic Studies*, 79, 307–339.
- Levinsohn, J., & Petrin, A. (2003). Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70, 317–341.
- McCaig, B., & Pavcnik, N. (2013). Moving out of Agriculture: Structural Change in Vietnam. *NBER Working Paper*, No. 19616.
- Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71, 1695–1725.
- Melitz, M. J., & Ottaviano, G. (2008). Market size, trade, and productivity. *Review of Economic Studies*, 75, 295–316.
- Melitz, M. J., & Redding, S. J. (2014). Missing gains from trade? *American Economic Review: Papers and Proceedings*, 2014(104), 317–321.
- Muendler, M.A. (2004). Trade, technology and productivity: A study of Brazilian manufacturers, 1986–1998. CESifo Working Paper Series 1148.
- Newman, C., Rand, J., Talbot, J., & Tarp, F. (2015). Technology transfers, foreign investment and productivity spillovers. *European Economic Review*, 76, 168–187.
- Newman, C., Rand, J., & Tarp, F. (2013). Industry switching in developing countries. *World Bank Economic Review*, 27, 357–388.
- Newman, C., Rand, J., Tarp, F., & Anh, N. T. T. (2016). Exporting and productivity: Learning from Vietnam. *Journal of African Economies*, 26(1), 67–92.
- Olley, G. S., & Pakes, A. (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, 64, 1263–1297.
- Pavcnik, N. (2002). Trade liberalization, exit, and productivity improvements: Evidence from Chilean plants. *Review of Economic Studies*, 69, 245–276.
- Shor, A. (2004). Heterogeneous productivity response to tariff reduction. Evidence from Brazilian manufacturing firms. *Journal of Development Economics*, 75, 373–396.
- Sutton, J. (1998). *Technology and market structure: Theory and History*. Cambridge: MIT Press.
- Syverson, C. (2011). What determines productivity? *Journal of Economic Literature*, 49, 326–365.
- Roberts, M. J., & Tybout, J. R. (1997). The decision to export in Colombia: An empirical model of entry with sunk costs. *American Economic Review*, 87, 545–564.
- Topalova, P., & Khandelwal, A. (2011). Trade liberalization and firm productivity: The case of India. *Review of Economics and Statistics*, 93(3), 995–1009.
- Tybout, J. R., De Melo, J., & Corbo, V. (1991). The effects of trade reforms on scale and technical efficiency: New evidence from Chile. *Journal of International Economics*, 31, 557–586.
- Van Biesebroeck, J. (2008). The sensitivity of productivity estimates: Revisiting three important debates. *Journal of Business and Economic Statistics*, 26(3), 311–328.
- Van Biesebroeck, J. (2005). Exporting raises productivity in sub-Saharan African manufacturing firms. *Journal of International Economics*, 67, 373–391.